UG (NEP) 8th SEMESTER
COMPUTER APPLICATIONS: SOFTWARE ENGINEERING
(CAP822J1)

Session: June 2026 No. of Questions: 09
Time Allowed: 2:00 Hours Max. Marks: 75 Min. Marks: 27

NOTE: Attempt all questions from Section “A” and “B” and only two questions from Section “C”.

Section A: Short Answer Type Questions (8 × 1.5 = 12 Marks)

Answer all questions in a maximum of 20 words each.

i. What is meant by Software Crisis?
ii. What is the main objective of Software Engineering?
iii. What is the difference between functional and non-functional requirements?
iv. What is the purpose of the COCOMO model in software engineering?
v. Define modularity in software design engineering.
vi. What is meant by software testing?
vii. What are the main challenges in software engineering?
viii. What is Component-based development?

Section B: Medium Answer Type Questions (4 × 7.5 = 30 Marks)

Answer all the four questions with internal choice in a maximum of 150 words each.

2. Explain the Waterfall Model of software development with advantages and disadvantages.

OR

Differentiate between Incremental Model and Iterative Model in software development.

3. Explain the Requirement Elicitation Process and discuss any two requirement elicitation techniques.

OR

Differentiate between Size-Oriented Metrics and Function-Oriented Metrics used in software measurement.

4. Explain the concepts of coupling and cohesion with suitable examples.

OR

Describe the different types of software testing techniques used in software development.

5. Explain Aspect Oriented development in detail.

OR

What is the difference between Black-Box Testing and White-Box Testing?

Section C: Long Answer Type Questions (2 × 16.5 = 33 Marks)

Answer only two questions in a maximum of 400 words each.

6. Explain the Agile Process Model Development in software engineering. Discuss its features, advantages, disadvantages, and applications in software development.
7. Explain the Capability Maturity Model Integration (CMMI) Model in software engineering. Discuss its features, advantages, and role in software cost estimation and project management.
8. Explain the basics of Design Engineering. Also explain the role of structured design methodology in software development.
9. Write detailed notes on Functional Independence, refinement and Refactoring.